

Combing hedgehogs over a field

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8 September 2026

Hedgehog theorem

Theorem (Poincaré 1885)

There is no non-vanishing vector field on $\{x^2 + y^2 + z^2 = 1\} \subseteq \mathbb{R}^3$.

Vector field = section of the tangent bundle T_{S^2} .

$$0 \rightarrow T_{S^2} \rightarrow S^2 \times \mathbb{R}^3 \xrightarrow{\nu} S^2 \times \mathbb{R} \rightarrow 0$$

$$\nu((x, y, z), (v_1, v_2, v_3)) = ((x, y, z), xv_1 + yv_2 + zv_3)$$

Theorem: There are no cont. functions f_1, f_2, f_3 on S^2 such that
(i) f_1, f_2, f_3 are jointly non-vanishing and (ii) $xf_1 + yf_2 + zf_3 = 0$.

What if we change

- $\mathbb{R}$ to an arbitrary field k ? ($\text{char } k \neq 2$)
- $\{x^2 + y^2 + z^2 = 1\}$ to $\{q = 1\}$ for a quadratic form q ?

Algebraic sphere

Notation

- $S_k^2 := \{x^2 + y^2 + z^2 = 1\} \subseteq \mathbb{A}_k^3$,
- $R = k[x, y, z]/(x^2 + y^2 + z^2 - 1)$ ring of regular functions,
- Tangent bundle is given by the projective module T

$$0 \rightarrow T \rightarrow R^3 \xrightarrow{\nu} R \rightarrow 0$$

$$\nu(v_1, v_2, v_3) = (xv_1 + yv_2 + zv_3)$$

Note that ν splits via

$$s(v) = (xv, yv, zv), \quad \nu s(v) = (x^2 + y^2 + z^2)v = v,$$

so

$$R^3 = T \oplus R \cdot (x, y, z).$$

Question

Do there exist $f_1, f_2, f_3 \in R$ such that

$$(i) Rf_1 + Rf_2 + Rf_3 = R, \quad (ii) xf_1 + yf_2 + zf_3 = 0?$$

Algebraic sphere (cont.)

Question

Do there exist $f_1, f_2, f_3 \in R = k[x, y, z]/(x^2 + y^2 + z^2 - 1)$ s.t.

$$(i) Rf_1 + Rf_2 + Rf_3 = R, \quad (ii) xf_1 + yf_2 + zf_3 = 0?$$

Answer

- $k = \mathbb{R}$ no
- $k = \mathbb{C}$ yes: $f_1 = -z, f_2 = -iz, f_3 = x + iy$
- $k = \mathbb{Q}_p, p \neq 2$ yes
- $k = \mathbb{Q}_2$??? (Umberto Zannier '20)
- $k = \mathbb{Q}_2$ yes (A.-Levine, 23)
- general k : yes if and only if $x_1^2 + x_2^2 + x_3^2 + x_4^2 = -1$ has a solution in k (A.-Levine, 23)

Algebraic sphere (cont.)

Question

Do there exist $f_1, f_2, f_3 \in R = k[x, y, z]/(x^2 + y^2 + z^2 - 1)$ s.t.

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Unimodular and completable rows

Definition

R commutative ring, $u = (u_1, u_2, \dots, u_n) \in R^n$ is

- *unimodular* if $Ru_1 + Ru_2 + \dots + Ru_n = R$,
- *completable over R* if u may be completed to a basis of R^n ,
($\Leftrightarrow \exists g \in \mathrm{GL}_n(R)$ with $(u_1, u_2, \dots, u_n)$ being the first row).

Easy: completable $\Rightarrow$ unimodular

Example: (x, y, z) is unimodular over $k[x, y, z]/(x^2 + y^2 + z^2 - 1)$.

Lemma

$R = k[x, y, z]/(x^2 + y^2 + z^2 - 1)$. The following are equivalent:

- 1 $\exists f = (f_1, f_2, f_3) \in T \subseteq R^3$ unimodular,
- 2 $T \cong R \oplus L$ for some L ,
- 3 $T \cong R^2$,
- 4 (x, y, z) is completable.

Proof: $1 \Leftrightarrow 2$: $T = Rf \oplus L$; $2 \Leftrightarrow 3 \Leftrightarrow 4$: $R^3 = T \oplus R \cdot (x, y, z)$.

Completions of unimodular rows

$$k = \mathbb{C}$$

$$M = \begin{pmatrix} x & y & z \\ 1 & i & 0 \\ \frac{iz}{2} & \frac{z}{2} & -y - ix \end{pmatrix}, \quad M^{-1} = \begin{pmatrix} x - iy & y(ix + y) + \frac{z^2}{2} & -iz \\ ix + y & -x(ix + y) - \frac{iz^2}{2} & z \\ z & -\frac{z(x - iy)}{2} & ix - y \end{pmatrix}$$

$$k = \mathbb{Q}_2, \omega = \sqrt{-7} \text{ (Müller '26; earlier example in '24)}$$

$$M = \begin{pmatrix} x & y & z \\ \omega y - 5z & 2\omega x - \omega y + 8z + 3 & -5x + 5y + 4\omega z + \omega \\ -6z + 1 & 3\omega x + \omega y + 9z + 1 & -7x + y + 5\omega z \end{pmatrix}$$
$$\det M = 5.$$

Remark

Müller '26 gives an explicit formula for completion starting from $x_1, x_2, x_3, x_4 \in k$ satisfying $x_1^2 + x_2^2 + x_3^2 + x_4^2 = -1$.

Notation

- $q = a_1x_1^2 + a_2x_2^2 + \dots + a_{n+1}x_{n+1}^2$, $a_i \in k^\times$, quadratic form,
- $Q^\circ := \{q = 1\} \subseteq \mathbb{A}^{n+1}$ smooth affine quadric,
- $R = k[x_1, x_2, \dots, x_{n+1}]/(q - 1)$ ring of regular functions,
- Tangent bundle is given by the projective module T

$$0 \rightarrow T \rightarrow R^{n+1} \xrightarrow{\nu} R \rightarrow 0$$

$$\nu(v_1, v_2, \dots, v_{n+1}) = (a_1x_1v_1 + a_2x_2v_2 + \dots + a_{n+1}x_{n+1}v_{n+1})$$

Note that ν splits via $s(v) = (x_1v, x_2v, \dots, x_{n+1}v)$, so

$$R^{n+1} = T \oplus R \cdot (x_1, x_2, \dots, x_{n+1}).$$

Questions

- 1 $\exists?$ unimodular $(f_1, \dots, f_{n+1}) \in T \subseteq R^{n+1}$? ($\Leftrightarrow T \cong R \oplus T'$)
- 2 Is $(x_1, \dots, x_{n+1}) \in R^{n+1}$ completable? ($\Leftrightarrow T \cong R^n$)

Remark: $1 \Leftrightarrow 2$ for $n = 1, 2$.

(Non-)existence of non-vanishing sections

Notation

- $D(q) \subseteq k^\times$ set of non-zero values of q ,
- $D(q)^2 := \{ab \mid a, b \in D(q)\}$,
- $[D(q)], [D(q)^2] \leq k^\times$ subgroups generated by $D(q)$ and $D(q)^2$

Theorem (A.-Levine '23)

Let k be a perfect field of characteristic $\neq 2$.

- 1 If n is odd, then $T \cong R \oplus T'$.
- 2 If $n > 0$ is even and $T \cong R \oplus T'$ then $-\prod_{i=1}^{n+1} a_i \in [D(q)^2]$.
- 3 If $n > 0$ is even, $1 \in D(q)$ and $-1 \in [D(q)]$ then $T \cong R \oplus T'$.

If $1 \in D(q)$ then $[D(q)^2] = [D(q)]$ and

$$-\prod_{i=1}^{n+1} a_i \in [D(q)^2] \Leftrightarrow -1 \in [D(q)].$$

$n = 1$

- $R = k[x_1, x_2]/(a_1x_1^2 + a_2x_2^2 - 1)$
- $T = \{(v_1, v_2) \in R^2 \mid a_1x_1v_1 + a_2x_2v_2 = 0\} \subseteq R^2$
- $(a_2x_2, -a_1x_1) \in T$ is unimodular

$n = 2m - 1$

- $R = k[x_1, x_2, \dots, x_{2m}]/(a_1x_1^2 + a_2x_2^2 + \dots + a_{2m}x_{2m}^2 - 1)$
- $T = \{(v_1, \dots, v_{2m}) \in R^{2m} \mid \sum_{i=1}^{2m} a_ix_iv_i = 0\} \subseteq R^{2m}$
- $(a_2x_2, -a_1x_1, a_4x_4, -a_3x_3, \dots, a_{2m}x_{2m}, -a_{2m-1}x_{2m-1}) \in T$ is unimodular

$n = 1$

- $R = k[x_1, x_2]/(a_1x_1^2 + a_2x_2^2 - 1)$
- $T = \{(v_1, v_2) \in R^2 \mid a_1x_1v_1 + a_2x_2v_2 = 0\} \subseteq R^2$
- $(a_2x_2, -a_1x_1) \in T$ is unimodular

$n = 2m - 1$

- $R = k[x_1, x_2, \dots, x_{2m}]/(a_1x_1^2 + a_2x_2^2 + \dots + a_{2m}x_{2m}^2 - 1)$
- $T = \{(v_1, \dots, v_{2m}) \in R^{2m} \mid \sum_{i=1}^{2m} a_ix_iv_i = 0\} \subseteq R^{2m}$
- $(a_2x_2, -a_1x_1, a_4x_4, -a_3x_3, \dots, a_{2m}x_{2m}, -a_{2m-1}x_{2m-1}) \in T$ is unimodular

Pfister neighbors

Pfister neighbor

- $\langle a_1, a_2, \dots, a_n \rangle := \sum_{i=1}^n a_i x_i^2$
- $\langle\langle a_1, a_2, \dots, a_N \rangle\rangle := \bigotimes_{i=1}^N \langle 1, -a_i \rangle$ is an N -fold Pfister form
Example: $\langle 1, -a, -b, ab \rangle = x^2 - ay^2 - bz^2 + abt^2$.
- q is a Pfister neighbor if $\exists a \in k^\times$ such that $\langle a \rangle \cdot q$ is a subform of an N -fold Pfister form with $2^{N-1} < \dim q$
Example: $\langle a_1^{-1} \rangle \langle a_1, a_2, a_3 \rangle \subseteq \langle 1, \frac{a_2}{a_1}, \frac{a_3}{a_1}, \frac{a_2 a_3}{a_1^2} \rangle = \langle\langle -\frac{a_2}{a_1}, -\frac{a_3}{a_1} \rangle\rangle$
Example: $\sum_{i=1}^{n+1} x_i^2 \subseteq \langle 1, 1 \rangle^{\otimes N}$, where $2^{N-1} < n+1 \leq 2^N$.

Lemma

If q is a Pfister neighbor of ϕ then $[D(q)^2] = D(\phi)$.

Combing hedgehogs in higher dimension

Corollary

$q = \sum_{i=1}^{n+1} x_i^2$, $n \geq 1$. Denote $N = \lfloor \log_2(n+1) \rfloor + 1$.

Then $T \cong R \oplus T'$ if and only if one of the following holds:

- ① n is odd,
- ② n is even and $\sum_{i=1}^{2^N} y_i^2 = -1$ has a solution in k .

Proof.

2) $1 \in D(q)$ hence $T \cong R \oplus T' \Leftrightarrow -1 \in [D(q)] = D(\phi)$ for the Pfister form $\phi := \sum_{i=1}^{2^N} y_i^2$. □

Remark

Min. N such that $\sum_{i=1}^N y_i^2 = -1$ has a solution in k is *Stufe* $s(k)$. It is always a power of 2 or infinite. $s(\mathbb{R}) = \infty$, $s(\mathbb{C}) = 1$, $s(\mathbb{Q}_p) = 1$ for $p = 4m + 1$, $s(\mathbb{Q}_p) = 2$ for $p = 4m + 3$, $s(\mathbb{Q}_2) = 4$.

Combing hedgehogs in dim 2

Theorem (A.-Fasel '26)

Let $q = a_1x^2 + a_2y^2 + a_3z^2 \in k[x, y, z]$ and denote

$$a := -a_2a_3, \quad b := -a_1a_3, \quad \Delta := -a_1a_2a_3, \quad R := k[x, y, z]/(q - 1).$$

Then the following are equivalent:

- ① $\Delta \in [D(q)^2]$,
- ② $x_0^2 - ax_1^2 - bx_2^2 + abx_3^2 = \Delta$ has a solution over k ,
- ③ the row (x, y, z) is completable over R ,
- ④ $T \cong R^2$,
- ⑤ $T \cong R \oplus L$.

Proof.

$1 \Leftrightarrow 2$ Pfister neighbor; $3 \Leftrightarrow 4 \Leftrightarrow 5$ as above; $5 \Rightarrow 1$ theorem with Levine; $2 \Rightarrow 3$ explicit construction (used AI). □

Examples in dimension 2

$$q = -x^2 + ay^2 + az^2, R = k[x, y, z]/(1 + x^2 - ay^2 - az^2)$$

Scaling followed by an elementary transformation

$$(x, y, z) \mapsto (2x + ay^2 + az^2, y, z) = ((x + 1)^2, y, z).$$

The row $(x + 1, y, z)$ is unimodular with

$$\frac{1}{2}(1 - x) \cdot (x + 1) + \frac{1}{2}ay \cdot y + \frac{1}{2}az \cdot z = 1,$$

thus $((x + 1)^2, y, z)$ may be completed (Swan-Towber '75) as

$$\begin{pmatrix} (x + 1)^2 & y & z \\ -y - a(x + 1)z & \frac{a^2 z^2}{4} & \frac{2 - 2x - a^2 yz}{4} \\ -z + a(x + 1)y & \frac{2x - 2 - a^2 yz}{4} & \frac{a^2 y^2}{4} \end{pmatrix}$$

Transforming back we obtain

$$\begin{pmatrix} x & y & z \\ -\frac{2y + a(x+3)z}{4} & \frac{a^2 z^2}{4} & \frac{2 - 2x - a^2 yz}{4} \\ -\frac{2z + a(x+3)y}{4} & \frac{2x - 2 - a^2 yz}{4} & \frac{a^2 y^2}{4} \end{pmatrix}.$$

Examples in dimension 2 (cont.)

$$q = ax^2 + by^2 - abz^2, R = k[x, y, z]/(1 - ax^2 - by^2 + abz^2)$$

Consider the quaternion algebra $H := (a, b)_k$ and denote

$$H^0(R) := Ri \oplus Rj \oplus Rij \subseteq R \otimes_k H, \quad \rho := xi + yj + zij \in H^0(R).$$

Then decomposition $R^3 = T \oplus R(x, y, z)$ becomes

$$H^0(R) = T \oplus R\rho, \quad T := (R\rho)^\perp = \{h \in H^0(R) \mid \text{tr}(\rho h) = 0\}$$

We have $\rho^2 = 1$ and $\rho T = T$ hence

$$T = T_+ \oplus T_-, \quad T_+ := \frac{1+\rho}{2} T, \quad T_- := \frac{1-\rho}{2} T$$

and $T_+ \cong T_-^\vee$. Consider the elements

$$\begin{aligned} e_1^+ &:= \frac{1+\rho}{2}(i - ax\rho), & e_2^+ &:= \frac{1+\rho}{2}(j - by\rho), \\ e_1^- &:= \frac{1-\rho}{2}(i - ax\rho), & e_2^- &:= \frac{1-\rho}{2}(j - by\rho). \end{aligned}$$

One can find a basis for T as $u := e_1^+ + \lambda_1 e_1^- + \mu_1 e_2^-$,
 $v := e_2^+ + \lambda_2 e_1^- + \mu_2 e_2^-$ with $\lambda_1 = \mu_2 = abz$, $\mu_1 = 2a$, $\lambda_2 = -2b$.

Completing in higher dimension I

Theorem (Revoy '79)

If q is isotropic (i.e. $q = 0$ has a nontrivial solution) then $T \cong R^n$.

Proof.

Change coordinates so that $q = x_1x_2 + \sum_{i=3}^{n+1} b_i x_i^2$, and then an explicit sparse matrix over $R = k[x_1, \dots, x_{n+1}]/(q - 1)$.

E.g. for $n = 2$ the row $v := (-2b_3x_3, 0, x_2)$ is unimodular and

$$v \in T = \{(v_1, v_2, v_3) \in R^3 \mid x_2v_1 + x_1v_2 + 2b_3x_3v_3 = 0\} \subseteq R^3.$$



Definition

- *Witt index* $i(q)$ is the dimension of a maximal totally isotropic subspace of q .
- *First Witt index* $i_1(q)$ of an anisotropic q is the minimal positive Witt index $i(q_F)$ as field extension F/k varies.

Example

- q an anisotropic Pfister form, $\dim q = 2^N$, then $i_1(q) = 2^{N-1}$.
- More generally, if q is an anisotropic Pfister neighbor with $\dim q = 2^{N-1} + m$, $1 \leq m \leq 2^{N-1}$, then $i_1(q) = m$.

Completing in higher dimension II

Theorem (A.-Fasel '26)

Let $q = \sum_{i=1}^{n+1} a_i x_i^2$, $R = k[x_1, \dots, x_{n+1}]/(q-1)$ and $\tilde{q} = -x_0^2 + q$. Suppose $\tilde{q}$ is anisotropic and $i_1(\tilde{q}) \geq 2$.

Then every rank n stably trivial projective module over R is trivial. In particular, $T \cong R^n$.

Proof.

$Q := \{\tilde{q} = 0\} \subseteq \mathbb{P}^{n+1}$, $Q^\infty := Q \cap \{x_0 = 0\}$, $Q^\circ := Q \setminus Q^\infty \cong \operatorname{Spec} R$

By the assumption Q^∞ is stably birational to Q , so $H^n(Q^\circ, \mathcal{A}) = 0$ for every strictly $\mathbb{A}^1$ -invariant sheaf, all the obstructions vanish. $\square$

Example

$q = ax^2 + by^2 - abz^2$ anisotropic, $\tilde{q} = -\langle\langle a, b \rangle\rangle$ has $i_1(\tilde{q}) = 2$.

Definition

- X a smooth manifold, E a vector bundle over X ,
- suppose X connected, compact and orientable, E orientable,
- suppose $\dim X = \text{rank } E = n$.

Euler class: $e^{\text{top}}(E) \in H^n(X, \mathbb{Z}) \cong H_0(X, \mathbb{Z})$.

$$e^{\text{top}}(E) = [s_0(X) \cap s(X)] = \sum_{x \in s_0(X) \cap s(X)} \pm [x] \in H_0(X, \mathbb{Z}),$$

where $s_0: X \rightarrow E$ zero section and $s: X \rightarrow E$ general section.

Theorem

$E \cong 1_X \oplus E'$ if and only if $e^{\text{top}}(E) = 0$.

Theorem (Gauß-Bonnet formula)

$e^{\text{top}}(T_X) = \chi(X) \in \mathbb{Z} \cong H_0(X, \mathbb{Z})$.

Chow-Witt zero cycles

Grothendieck-Witt ring of a field F

$$\mathrm{GW}(F) := \left(\bigoplus_{\phi} \mathbb{Z}[\phi] \right) / ([\phi_1 \perp \phi_2] - [\phi_1] - [\phi_2])_{\phi_1, \phi_2},$$

ϕ, ϕ_1 , and ϕ_2 are nondegenerate quadratic forms over F , $[\phi]$ is the isometry class of ϕ .

Chow-Witt zero cycles (Barge-Morel '00)

X a smooth variety over k ; X_0, X_1 points of dimension 0 and 1.

$$\widetilde{\mathrm{CH}}_0(X) := \mathrm{Coker} \left(\bigoplus_{y \in X_1} K_1^{\mathrm{MW}}(k(y)) \xrightarrow{\partial} \bigoplus_{x \in X_0} \mathrm{GW}(k(x)) \right).$$

If k is alg. closed then

$$\widetilde{\mathrm{CH}}_0(X) \cong \mathrm{CH}_0(X) = \mathrm{Coker} \left(\bigoplus_{y \in X_1} k(y)^{\times} \xrightarrow{\partial} \bigoplus_{x \in X_0} \mathbb{Z} \right).$$

Motivic Euler class

Definition

- X a smooth variety, E a vector bundle over X ,
- suppose $\det E \cong \omega_X^*$ and $\dim X = \operatorname{rank} E$.

Euler class: $e^{\operatorname{mot}}(E) \in \widetilde{\operatorname{CH}}_0(X)$.

$$e^{\operatorname{mot}}(E) = [s_0(X) \cap s(X)] = \sum_{x \in s_0(X) \cap s(X)} \langle \lambda_x \rangle \in \widetilde{\operatorname{CH}}_0(X),$$

$s_0: X \rightarrow E$ zero section and $s: X \rightarrow E$ section transversal to s_0 .

Theorem (Morel '12, Asok-Fasel '16, Asok-Hoyois-Wendt '17)

Let X, E be as above and assume X is affine.

Then $E \cong \mathcal{O}_X \oplus E'$ if and only if $e^{\operatorname{mot}}(E) = 0$ in $\widetilde{\operatorname{CH}}_0(X)$.

Problem: $\widetilde{\operatorname{CH}}_0(Q^\circ)$ is hard to compute for $Q^\circ = \operatorname{Spec} R$ a smooth affine quadric.

Idea of the proof of A.-Levine theorem

$$q = \sum_{i=1}^{n+1} a_i x_i^2, \quad \tilde{q} := -x_0^2 + q, \quad Q := \{\tilde{q} = 0\} \subseteq \mathbb{P}^{n+1},$$

$$Q^\infty := Q \cap \{x_0 = 0\} = \{q = 0\} \subseteq \mathbb{P}^n, \quad Q^\circ := Q \setminus Q^\infty \cong \operatorname{Spec} R$$

There is a commutative diagram with an exact row

$$\begin{array}{ccccc} \widetilde{\operatorname{CH}}_0(Q^\infty) & \xrightarrow{i_*} & \widetilde{\operatorname{CH}}_0(Q) & \xrightarrow{j^*} & \widetilde{\operatorname{CH}}_0(Q^\circ) \longrightarrow 0 \\ & \searrow p_*^\infty & \downarrow p_* & & \\ & & \operatorname{GW}(k) & & \end{array}$$

p_*^∞, p_* are induced by Scharlau transfers $\operatorname{GW}(k(x)) \rightarrow \operatorname{GW}(k)$:

$$[\phi: V \times V \rightarrow k(x)] \mapsto [\phi: V \times V \rightarrow k(x) \xrightarrow{\operatorname{trace}} k]$$

with V viewed as a vector space over $k(x)$ on the left and over k on the right.

Idea of the proof of A.-Levine theorem (cont.)

$$\begin{array}{ccccc} \widetilde{\mathrm{CH}}_0(Q^\infty) & \xrightarrow{i_*} & \widetilde{\mathrm{CH}}_0(Q) & \xrightarrow{j_*} & \widetilde{\mathrm{CH}}_0(Q^\circ) \longrightarrow 0 \\ & \searrow p_*^\infty & \downarrow p_* & & \\ & & \mathrm{GW}(k) & & \end{array}$$

- $j^*(e^{\mathrm{mot}}(T_Q)) = e^{\mathrm{mot}}(T_{Q^\circ})$
- $p_*(e^{\mathrm{mot}}(T_Q)) = \frac{n}{2}\langle 1, -1 \rangle + \langle 2, 2 \prod_{i=1}^{n+1} a_i \rangle$ (n even)
(motivic Gauß-Bonnet, Levine-Raksit '20)
- Lemma: $\langle a, b \rangle \in p_*^\infty(\widetilde{\mathrm{CH}}_0(Q^\infty))$ if and only if $-ab \in [D(q)^2]$
This yields the necessary condition
- If $1 \in D(q)$ then Q is rational and p_* is an isomorphism
(stable birational invariance)
This yields the sufficient condition

Results and questions

$$q = \sum_{i=1}^{n+1} a_i x_i^2, \quad R = k[x_1, \dots, x_{n+1}]/(q - 1)$$

Theorem (A.-Levine '23)

- 1 If n is odd, then $T \cong R \oplus T'$.
- 2 If $n > 0$ is even and $T \cong R \oplus T'$ then $-\prod_{i=1}^{n+1} a_i \in [D(q)^2]$.
- 3 If $n > 0$ is even, $1 \in D(q)$ and $-1 \in [D(q)]$ then $T \cong R \oplus T'$.

Theorem (A.-Fasel '26)

Let $\tilde{q} = -x_0^2 + q$ and suppose $\tilde{q}$ is anisotropic and $i_1(\tilde{q}) \geq 2$.
Then every rank n stably trivial projective module over R is trivial.
In particular, $T \cong R^n$.

- For $n \geq 4$ is $-\prod_{i=1}^{n+1} a_i \in [D(q)^2]$ sufficient for $T \cong R \oplus T'$?
- Explicit formulas in higher dimension.

Thank you!